

Cybersecurity Incident Report

One potential explanation for the website's connection timeout error message is that the server got overwhelmed by a flood for example (a DoS attack) and could not respond to the requests it received.

The logs show that the server receives multiple SYN packet requests from a persistent IP address (203.0.113.0), the server responds and establishes connection at first while also handling other client requests, then struggles to handle those later on, until completely unable to respond.

This event could be that the malicious IP address belongs to an attacker using a SYN flood attack on the server carry on a DoS attack.

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. SYN : client requests to establish a connection (SYNCHRONISATION) with the server.
2. SYN, ACK : server responds and acknowledges the establishment of the connection while also requesting an acknowledgement back from the client.
3. ACK : finalizes the connection establishment process as both parties acknowledged the connection being there.

As a malicious actor sends a large number of SYN packets all at once, the server gets overwhelmed once the number of requests is more than what it can handle, it then stops being able to respond, causing the server's normal operations to be at a halt too, as it becomes unable to respond to legit requests too.

The logs indicate a SYN flood attack coming from an attacker with the IP address of 203.0.113.0, as stated before, the web server gets overwhelmed once it can't handle the number of requests anymore and the website stops responding and starts sending timeout error messages.